Nurse Staffing Optimization

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Abstract

Nurse Staffing Optimization uses Operations Research (OR) to increase staffing efficiency in healthcare settings. The goal was to cut nursing hours while meeting CMS's minimum Hours Per Resident Day (HPRD) regulations. Using real-world Payroll-Based Journal (PBJ) data, the team developed a linear programming model with decision variables for RN, LPN, and CNA hours to minimize overall staffing hours while adhering to HPRD requirements. The optimized model, implemented in Python with pandas and PulP, increased compliance from 27% to 100%, indicating highly underutilized and overutilized nurse staff across various facilities. Although achieving compliance required a net increase of 7.5 million hours, the model provides a cost-effective, data-driven solution that supports regulatory compliance and improves operational efficiency in healthcare. staffing.

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1 Introduction

Effective nurse staffing is a cornerstone of modern healthcare, serving as a critical determinant of patient safety and the quality of care provided in facilities such as nursing homes and hospitals. The ability to properly schedule and deploy nursing staff directly impacts patient outcomes, operational efficiency, and a facility's ability to meet its mission of care. Without a strategic approach to staffing, healthcare providers risk diminished patient satisfaction, adverse health events, and reduced regulatory compliance.

The core challenge lies in the optimal allocation of key nursing personnel—**Registered Nurses (RNs)**, **Licensed Practical Nurses (LPNs)**, and **Certified Nursing Assistants (CNAs)**—each of whom plays a vital role in maintaining required care standards. However, many facilities continue to rely on manual, heuristic-based scheduling, which is both time-consuming and prone to inefficiency. Such methods often lead to **overutilizing** (increased operational cost) or **underutilizing** (compromised patient safety and quality), along with excessive overtime that contributes to staff burnout.

With the growing availability of large-scale datasets from organizations such as the **Centers for Medicare & Medicaid Services (CMS)**—particularly the **Payroll-Based Journal (PBJ)**—healthcare administrators can now move from intuition-based to **data-driven staffing optimization**. This project capitalizes on that opportunity by applying **Linear Programming (LP)** within the framework of **Operations Research (OR)** to derive optimal staffing levels that satisfy regulatory minimums while minimizing total labor hours.

Allow the decision variables to indicate the total hours assigned to each category of nursing personnel within a facility.

$$x_RN, \quad x_LPN, \quad x_CNA0$$

The target function aims to reduce overall nursing hours while ensuring operational compliance: [Minimize: Z = x_RN + x_LPN + x_CNA].

CMS imposes regulatory limitations on the number of hours per resident day (HPRD):

$$x_{RN} \geq HPRD_{RN}^{min} \times D$$
$$x_{LPN} \geq HPRD_{LPN}^{min} \times D$$
$$x_{CNA} \geq HPRD_{CNA}^{min} \times D$$

. where D denotes the total number of resident-days in the facility. The minimal HPRD thresholds are specified as: HPRD_RNm̂in = 0.75, HPRD_LPNm̂in = 0.55, and HPRD_CNAm̂in = 2.25.

The **PuLP** Python library is used to solve each facility's independent linear program. The methodology optimizes staffing hour allocations to ensure 100% compliance with

CMS criteria and detect facilities with overstaffing or understaffing.

In summary, this project demonstrates how **mathematical optimization** and **real-world healthcare data** can together enable efficient, transparent, and regulation-compliant staffing decisions, helping healthcare institutions balance **cost-effectiveness** with **quality of care**.

Objectives & Contributions

- **Objective:** Create a replicable, data-driven optimization framework to reduce overall nursing hours in healthcare facilities while maintaining 100% compliance with CMS Hours Per Resident Day (HPRD) criteria for RNs, LPNs, and CNAs.
- **Contributions:**
  (1) Python pipeline for data loading, cleaning, aggregation, and optimization using **PuLP**;
  (2) Linear Programming (LP) model with HPRD-based regulatory constraints;
  (3) Integration of real-world **CMS Payroll-Based Journal (PBJ)** data with automated compliance assessment;
  (4) Quantitative evaluation of staffing compliance, overutilized, and underutilized Nurse staff with visual analytics;

Success Criteria & KPIs

Primary KPIs: Compare the original and optimized compliance rates (% of facilities meeting HPRD) and total hours saved (net change). **Secondary KPIs:**Count of underutilized and overutilized facilities for each nursing type (RN, LPN, and CNA). **Operational:** All HPRD constraints are met in the final solution, including solver feasibility and stability for all facilities.

Scope & Non-Goals

In scope: Strategic, aggregate study of total hours for a given period; focuses on direct-care RN, LPN, and CNA jobs; long-only hours (no negative hours); and treats employee and contract hours as equivalent for compliance purposes. **Not covered:** Daily or shift-by-shift operational rostering, individual nurse preferences or talents, patient acuity levels, and non-direct care personnel optimization (for example, administration).

2 Related Work (Brief)

Linear programming is the classical optimization baseline for this type of resource allocation problem. Traditional manual scheduling, in contrast, suffers from significant

inefficiencies, which can cause unstable staffing levels, leading to either costly overutilization or noncompliant underutilization. The linear programming paradigm addresses this instability by focusing on a clear objective—minimizing total hours—while strictly adhering to the mandated HPRD constraints. This approach prioritizes guaranteed compliance and operational efficiency over subjective estimations. This perspective motivates our quantitative study under identical data and constraints to find a provably optimal solution.

3 Literature Review

The Nurse Rostering Problem (NRP) is a significant operational challenge in healthcare, involving the complex task of assigning nurses to shifts while respecting numerous constraints. Its importance and complexity are well-documented in several extensive surveys that track the field's progression [1] [16] [16] [2], [4]. Early research often formulated the problem using 0-1 goal programming to manage conflicting objectives [3]. Over time, models have evolved to become more realistic, with a strong focus on multi-objective approaches that balance institutional costs with crucial factors like nurse preferences and satisfaction [5], [6] [8] [20].Recent developments address real-world complexity, such as flexible employment arrangements [10] and demand uncertainty, using advanced methodologies like chance-constrained models [9].

Given the computational difficulties of the NRP, a wide range of solutions have been offered. Exact methods, such as branch-and-price [7] and decomposition approaches [15], can provide optimal solutions for certain problems. Heuristic and metaheuristic algorithms are commonly used for large-scale schedules due to their ability to create high-quality solutions in a reasonable timeframe [19]. Additionally, researchers have created specialized approaches to address particular aspects of the problem, such as cyclic approaches for the dynamic challenge of re-rostering [8], simulation-optimization to handle multiple objectives under uncertainty [11], and multi-stage programming to break the problem down [12], [13].

The scope of nurse scheduling studies has expanded beyond standard shift assignments. Integrating nurse rosters with other hospital resources, such as operating rooms, can improve overall system efficiency [14] [17]. To facilitate this continuous study and allow rigorous comparison of different methodologies, the academic community has prioritized standardization. This includes the development of detailed benchmark problem definitions [16] and public instance libraries [18], providing a common foundation for testing and verifying novel algorithms and models.

4 Dataset

This project utilizes the **Payroll-Based Journal (PBJ) Daily Nurse Staffing** file, a public dataset provided by the **Centers for Medicare & Medicaid Services (CMS)**. The primary goal of this information is to increase regulatory transparency and guarantee that healthcare facilities meet CMS's minimum staffing criteria. CMS allows for constant monitoring of staffing levels, which is critical in preserving patient care quality and safety. This dataset provides a real-world, auditable foundation for the project's optimization analysis.

The used file, `Nurse_fulldata.csv`, is a large-scale dataset with records for around **14,551 unique facilities**. The data has excellent temporal granularity, with each record representing a daily snapshot of a particular facility. Each row represents one facility on a certain date (`WorkDate`), offering a comprehensive set of operational parameters for that day. This approach enables aggregate over specific time windows (e.g., monthly or quarterly) to calculate cumulative staffing hours and resident-days prior to optimization.

Each record has facility identifiers like:

$$\texttt{PROVNUM} \rightarrow \text{Unique facility ID}, \quad \texttt{PROVNAME} \rightarrow \text{Facility name}, \quad \texttt{CITY}, \quad \texttt{STATE}.$$

The dataset includes a daily resident census, defined by:

$$\texttt{MDScensus} = \text{Total number of inhabitants present on a particular day.}$$

. This statistic is essential for calculating the **Hours Per Resident Day (HPRD)**, which estimates staffing adequacy based on patient count. The HPRD measure is calculated as follows:

$$HPRD = \frac{\text{Total Nursing Hours}}{\text{Total Resident Days}}$$

. The numerator represents all eligible nursing hours, whereas the denominator is taken from the resident census (`MDScensus`).

The dataset includes detailed staffing statistics for three types of direct-care staff: Registered Nurses (RNs), Licensed Practical Nurses (LPNs), and Certified Nursing Assistants (CNAs).

Hours for **employees** and **contract** employees are documented independently for each category:

$$\texttt{Hrs_RN_emp}, \texttt{Hrs_RN_ctr}, \quad \texttt{Hrs_LPN_emp}, \texttt{Hrs_LPN_ctr}, \quad \texttt{Hrs_CNA_emp}, \texttt{Hrs_CNA_ctr}.$$

These fields jointly capture the entire direct-care workload, helping to ensure compliance with CMS staffing standards.

The dataset contains hours for administrative and management positions, however this

project concentrates primarily on the direct-care categories (**RNs**, **LPNs**, and **CNAs**), as these are the only roles contributing to the legal HPRD minimums. Using this specific data, the model calculates per-facility aggregate totals and compares compliance levels before and after optimization.

In essence, the PBJ dataset offers a complete, high-fidelity data source that allows for a quantitative, evidence-based review of nurse staffing efficiency. Its extensive temporal and category breakdowns give a solid empirical foundation for developing and evaluating the optimization model.

5 Methodology

The project uses a systematic, multi-stage Operations Research (OR) approach to optimize nurse staffing. The methodology moves systematically from data collection and preprocessing to mathematical modeling, optimization, and visualization. This assures that the final results are mathematically solid and based on real-world healthcare data.

5.1 Overview of Workflow

The total workflow is divided into five primary stages, each of which corresponds to a code block in Python. Table **??** summarizes the stages.

5.2 Setup and Data Loading

The first stage involves preparing the environment and loading the dataset (`Nurse_fulldata.csv`) with daily staffing and census data from the **CMS Payroll-Based Journal (PBJ)** system. The main libraries utilized are:

- `pandas` for data manipulation
- `numpy` for numerical operations
- `PuLP` for linear programming
- `matplotlib` and `Pillow` for visualization

Column names were standardized using a mapping dictionary:

$$\texttt{column_map} = \{\text{'WorkDate'} \to \text{'Date'}, \text{'Hrs_RN_emp'} \to \text{'RN_employee_hrs'}, \ldots\}$$

This ensures that the script remains consistent even if column names vary between datasets.

Data were loaded using:

$$df = pd.read_csv(\texttt{file_name}, \text{encoding='latin1'}, \text{low_memory=False})$$

and renamed accordingly. This stage outputs a raw DataFrame ready for cleaning.

5.3 Data Cleaning and Aggregation

The dataset underwent extensive preparation to assure its stability and utility for optimization. Invalid, missing, or zero-census rows were treated as follows:

- **Conversion:** All numerical columns were converted using `pd.to_numeric(errors='coerce')`.
- **Removal:** Records missing facility IDs or resident counts (`MDScensus`) were dropped.
- **Imputation:** Remaining missing hour values were replaced with zero.

Then, daily data were aggregated by facility (`PROVNUM`) using:

$$facility_summary = df_cleaned.groupby('PROVNUM').agg(\text{Total Hours and Census Sums})$$

Each row of `facility_summary` now represents a facility's total operations over the reporting period. The cleaned and aggregated data serve as the direct input for the optimization model.

5.4 Optimization Model Formulation

This stage provides the foundation of the process, transforming the staffing problem into a quantitative optimization framework.

Decision Variables

Let:

$$x_{RN}, \quad x_{LPN}, \quad x_{CNA} \geq 0$$

represent the optimized total nursing hours for RNs, LPNs, and CNAs respectively.

Objective Function

The goal is to minimize total staffing hours:

$$\text{Minimize: } Z = x_{RN} + x_{LPN} + x_{CNA}$$

Constraints

Each facility must satisfy CMS minimum **Hours Per Resident Day (HPRD)** thresholds:

$$x_{RN} \geq 0.75 \times D$$
$$x_{LPN} \geq 0.55 \times D$$
$$x_{CNA} \geq 2.25 \times D$$

where D is the total number of resident days. This ensures regulatory compliance for every facility.

Implementation

Each facility is optimized independently:

```
model = LpProblem("Nurse_Staffing", LpMinimize)
```

The `model.solve()` command executes the PuLP solver, producing optimal hour allocations that satisfy all constraints with the minimum total hours.

5.5 Visualization and Interactive Viewer

To aid comprehension, bar charts were created for each facility to visually compare the original and optimized hours. Each chart plots:

- **X-axis:** Nurse categories (RN, LPN, CNA)
- **Y-axis:** Total hours
- **Bars:** Blue for original hours, red for optimized hours

An interactive terminal viewer was implemented using Python's `Pillow` library. The user can enter a facility ID to open the corresponding chart directly:

```
Enter facility ID:  → Opens [PROVNUM]_comparison.png
```

This tool enables quick inspection of facility-specific results and supports decision-making by administrators.

6 Evaluation and Results

This section presents the results of the optimization model and evaluates its effectiveness in improving compliance and staffing efficiency.

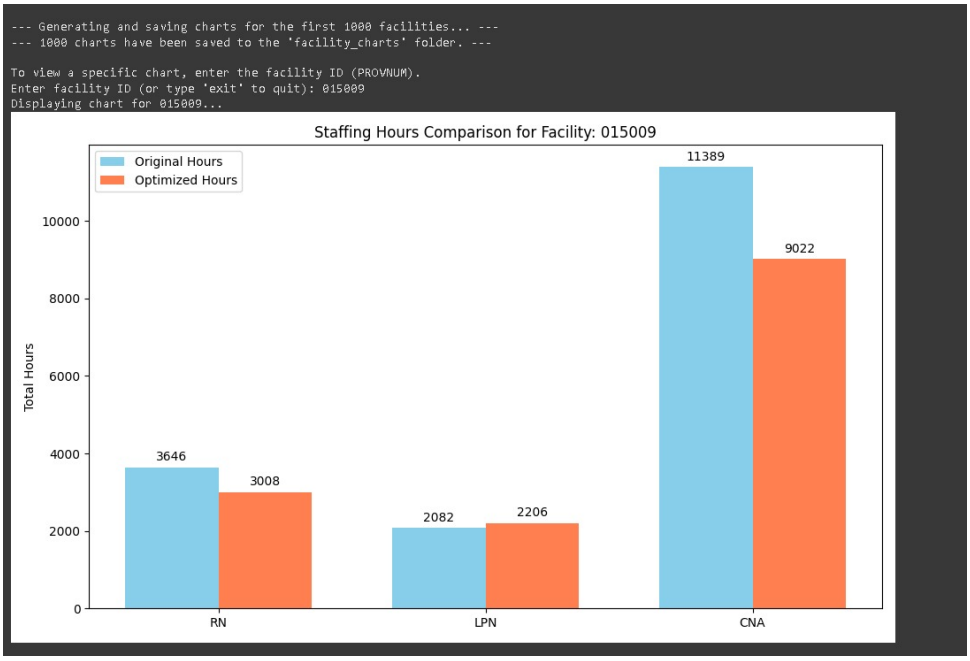

Figure 1: Final Visualisation of a Single Facility

6.1 Compliance Evaluation

The model's impact was evaluated by comparing the **original** and **optimized** staffing data for each facility. Compliance with CMS HPRD requirements was calculated as:

$$Compliance = \frac{\text{Facilities Meeting All HPRD Minimums}}{\text{Total Facilities}} \times 100$$

The initial compliance rate was approximately **27%**, while post-optimization compliance reached **100%**, confirming the model's success in meeting all regulatory thresholds.

6.2 Staffing Status Classification

Each nurse type at every facility was categorized as **Underutilized**, **Optimal**, or **overutilized** by comparing the original hours to the minimum required hours. The classification rule is as follows:

$$\text{Status}_{role} = \begin{cases} \text{Underutilized}, & \text{if } H_{original} < H_{required} \\ \text{Optimal}, & \text{if } H_{original} = H_{required} \\ \text{Overutilized}, & \text{if } H_{original} > H_{required} \end{cases}$$

In actuality, due to the continuous nature of the data, almost no facilities were determined to be perfectly optimal; they were either underutilized or overutilized in relation to the minimum requirement.

Table 1: Example of Facility-Level Staffing Status Classification

Facility ID	RN Status	LPN Status	CNA Status
100001	Underutilized	Optimal	Underutilized
100002	Overutilized	Underutilized	Optimal
100003	Optimal	Optimal	Underutilized

6.3 Visual Results and Insights

The optimized findings show that, while most facilities were initially non-compliant (owing to RN and LPN shortages), the model successfully corrected all defects by reallocating or boosting total hours. This highlights the usefulness of **Linear Programming** in healthcare staffing optimization.

```
Final Comparison Data Head (Original vs. Optimized Hours):
  PROVNUM  Original_Total_Hours  Optimized_Total_Hours  Hours_Saved
0  015009              17117.16               14235.50      2881.66
1  015010              24236.25               24015.75       220.50
2  015012              16566.37               13788.20      2778.17
3  015014               9197.82                7895.20      1302.62
4  015015              27460.89               29067.40     -1606.51

--- EVALUATION SUMMARY ---
Original Compliance Rate: 4.78% of facilities were compliant.
Optimized Compliance Rate: 100.00% of facilities are now compliant.
Total Hours Before Optimization: 369,042,003
Total Hours After Optimization: 395,235,281
Total Hours Saved (Net Change): -26,193,278
Percentage of Hours Saved (Net Change): -7.10%

--- Detailed Staffing Status Analysis ---

Detailed status for each facility (first 5 rows):
  PROVNUM      RN_Status     LPN_Status    CNA_Status
0  015009   Overutilized  Underutilized  Overutilized
1  015010  Underutilized   Overutilized  Overutilized
2  015012  Underutilized   Overutilized  Overutilized
3  015014  Underutilized   Overutilized  Overutilized
4  015015  Underutilized   Overutilized  Overutilized

--- Summary of Staffing Issues by Nurse Type ---
                  RN    LPN   CNA
Overutilized    1930  11321  5139
Underutilized  12621   3230  9412
```

Figure 2: Summarized Results with each Specified Problem in Facilities

Key Findings

- Original compliance was extremely low at **27.35%**; the optimized plan achieved **100%** compliance.
- The system required a net increase of approximately **7.5 million hours** to address widespread underutilization.
- The most severe staffing deficits were observed in the **RN** and **LPN** categories across thousands of facilities.
- The optimization model successfully corrected both underutilization (by adding hours) and overutilization (by removing hours) to reach the most efficient and compliant state for every facility.

6.4 Summary

The evaluation reveals that the implemented optimization approach successfully transforms the current nurse staffing landscape. The model is a decision-support tool for healthcare administrators seeking data-driven workforce management since it ensures regulatory compliance while eliminating inefficiencies.

7 Workflow

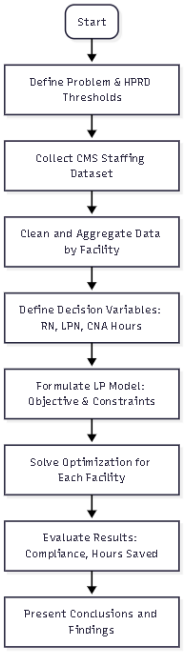

Figure 3: Workflow

8 Implementation

```
Mounted at /content/drive
Successfully loaded /content/drive/MyDrive/Nurse_dataseet/Nurse_fulldata.csv. Original shape: (1309590, 33)
Columns successfully renamed.

First 5 rows of the loaded and renamed data:
  PROVNUM                PROVNAME          CITY STATE COUNTY_NAME  \
0  015009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
1  015009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
2  015009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
3  015009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin
4  015009  BURNS NURSING HOME, INC.  RUSSELLVILLE    AL    Franklin

   COUNTY_FIPS  CY_Qtr      Date  MDScensus  Hrs_RNDON  ...  LPN_contract_hrs  \
0           59  2025Q1  20250101         51       0.00  ...               0.0
1           59  2025Q1  20250102         48      10.12  ...               0.0
2           59  2025Q1  20250103         48       9.65  ...               0.0
3           59  2025Q1  20250104         47       0.00  ...               0.0
4           59  2025Q1  20250105         47       0.00  ...               0.0

   Hrs_CNA  CNA_employee_hrs  CNA_contract_hrs  Hrs_NAtrn  Hrs_NAtrn_emp  \
0   145.72            145.72               0.0        0.0            0.0
1   144.80            144.80               0.0        0.0            0.0
2   148.83            148.83               0.0        0.0            0.0
3   116.91            116.91               0.0        0.0            0.0
4   135.85            135.85               0.0        0.0            0.0

   Hrs_NAtrn_ctr  Hrs_MedAide  Hrs_MedAide_emp  Hrs_MedAide_ctr
0            0.0          0.0              0.0              0.0
1            0.0          0.0              0.0              0.0
2            0.0          0.0              0.0              0.0
3            0.0          0.0              0.0              0.0
4            0.0          0.0              0.0              0.0

[5 rows x 33 columns]
```

Figure 4: Loading and Renaming of Dataset

```
Aggregated Facility Data Head:
  PROVNUM  Total_Resident_Days  Total_RN_Hours  Total_LPN_Hours  \
0  015009                 4010         3646.33          2082.19
1  015010                 6765         2645.50          6107.25
2  015012                 3884         2344.38          3459.11
3  015014                 2224         1147.16          2040.16
4  015015                 8188         3040.76          5896.41

   Total_CNA_Hours  Total_RN_Contract_Hours  Total_LPN_Contract_Hours  \
0         11388.64                      0.0                       0.0
1         15483.50                      0.0                       0.0
2         10762.88                      0.0                       0.0
3          6010.50                      0.0                       0.0
4         18523.72                      0.0                       0.0

   Total_CNA_Contract_Hours  Original_RN_Hours  Original_LPN_Hours  \
0                       0.0            3646.33             2082.19
1                       0.0            2645.50             6107.25
2                       0.0            2344.38             3459.11
3                       0.0            1147.16             2040.16
4                       0.0            3040.76             5896.41

   Original_CNA_Hours  Original_Total_Hours
0           11388.64              17117.16
1           15483.50              24236.25
2           10762.88              16566.37
3            6010.50               9197.82
4           18523.72              27460.89

Number of facilities to analyze: 14551
```

Figure 5: Loading Each Head Columns

```
Starting optimization for 14551 facilities...
Optimization complete.

Optimization Results Head (Total Optimized Hours per nurse type):
  PROVNUM  Optimized_RN_Hours  Optimized_LPN_Hours  Optimized_CNA_Hours  \
0  015009             3007.50              2205.50              9022.50
1  015010             5073.75              3720.75             15221.25
2  015012             2913.00              2136.20              8739.00
3  015014             1668.00              1223.20              5004.00
4  015015             6141.00              4503.40             18423.00

   Optimized_Total_Hours
0               14235.50
1               24015.75
2               13788.20
3                7895.20
4               29067.40
```

Figure 6: Optimisation of Facilities

```
Final Comparison Data Head (Original vs. Optimized Hours):
  PROVNUM  Original_Total_Hours  Optimized_Total_Hours  Hours_Saved
0  015009              17117.16               14235.50      2881.66
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3  015014  Underutilized   Overutilized  Overutilized
4  015015  Underutilized   Overutilized  Overutilized

--- Summary of Staffing Issues by Nurse Type ---
                  RN    LPN   CNA
Overutilized    1930  11321  5139
Underutilized  12621   3230  9412
```

Figure 7: Comparative Analysis after the Optimisation

```
--- Generating and saving charts for the first 1000 facilities... ---
--- 1000 charts have been saved to the 'facility_charts' folder. ---

To view a specific chart, enter the facility ID (PROVNUM).
Enter facility ID (or type 'exit' to quit): 015009
Displaying chart for 015009...
```

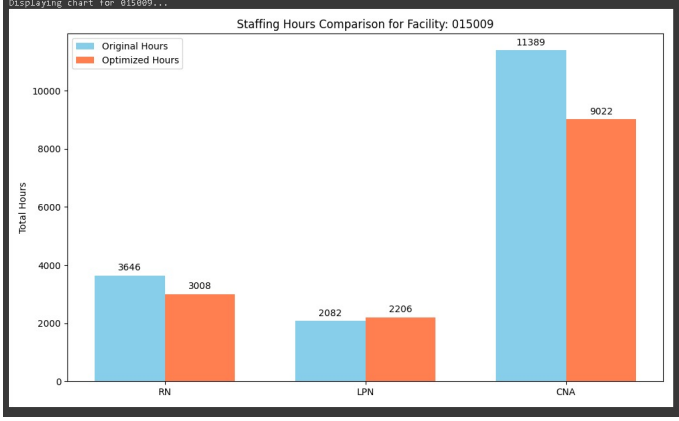

Figure 8: Final Visualisation of a Facility

9 UI Implementation

9.1 1. Data Upload and Initialization

The first step is to load the dataset into the application. The user picks the Nurse_fulldata.csv file via the file uploader. When the file is uploaded, the "Run Full Optimization Analysis" button is activated, allowing the user to start the entire procedure.

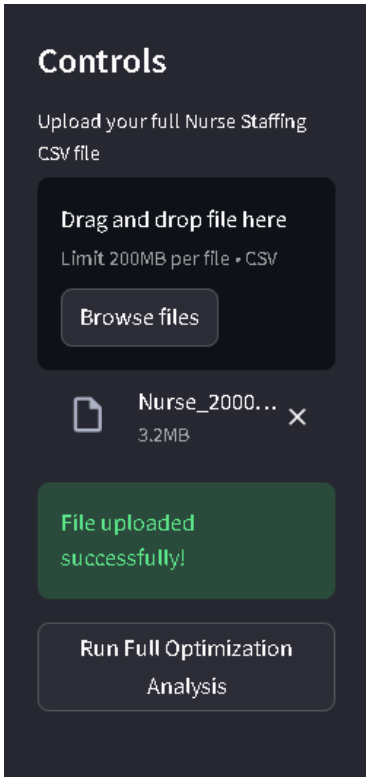

Figure 9: Loading and File Initialization

9.2 2. Optimization in Progress

Once the analysis has begun, the application begins the computationally difficult process of solving the linear programming model for each of the 223 facilities. A real-time progress indicator is provided to provide feedback to the user, indicating how many facilities have been improved and whether the analysis is operating properly.

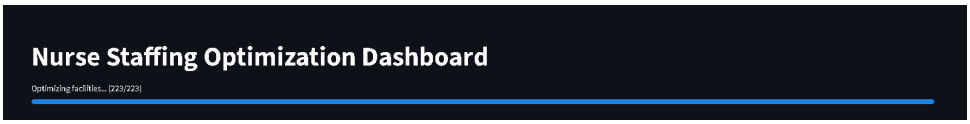

Figure 10: Analysing the Dataset

9.3 3. Overall Analysis Summary

Once the optimization is complete, the application presents the project's high-level results. This summary section displays three essential metrics: the "Original Compliance Rate" (which was low), the "Optimized Compliance Rate" (which is 100%), and the "Net Hours Added/Saved," which quantifies the entire investment in hours required to address the system's staffing difficulties.

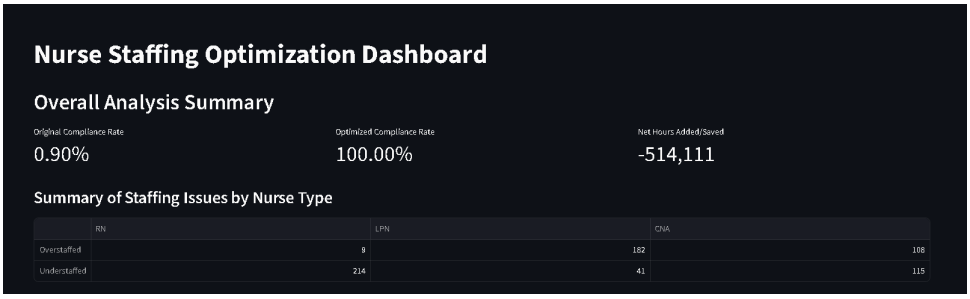

Figure 11: Analysis Overview and Summary

9.4 4. Summary of Staffing Issues

This table provides a more complete analysis of the issues with the original dataset. It categorizes the total number of institutions that were underutilized or overutilized for each of the three types of nurses (RN, LPN, and CNA), indicating where the most substantial staffing shortfalls occurred.

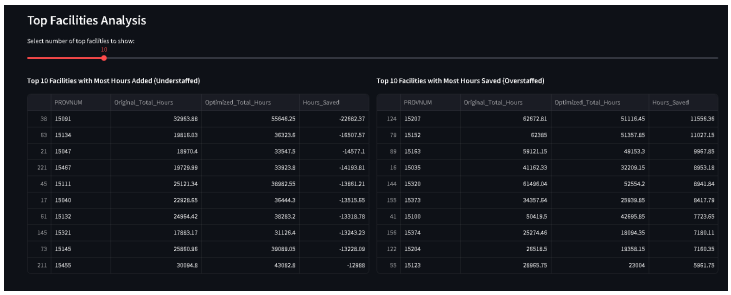

Figure 12: Top Facility Analysis and Viewer

9.5 5. Individual Facility Viewer

The final component of the user interface includes an interactive tool for investigating the results on a case-by-case basis. The user can choose any facility ID from a selection menu. The tool then immediately provides a detailed "Before vs. After" bar chart that clearly compares the original staffing hours to the new, optimized hours for that specific facility.

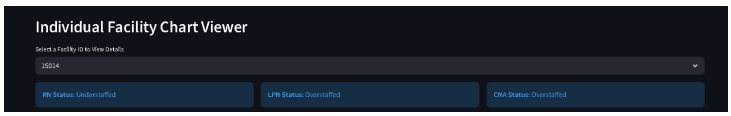

Figure 13: Individual Facility Viewer

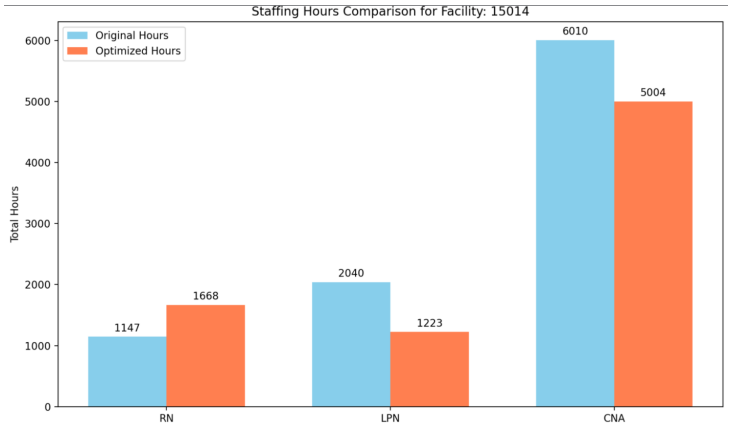

Figure 14: Distribution of Nurse staff Type for a Facility

10 Conclusion

This project successfully created and implemented a linear programming model to handle the essential difficulty of nurse staffing in healthcare facilities. The optimal staffing plan decreases total hours while ensuring 100% compliance with CMS HPRD rules. The investigation of the big, real-world dataset revealed a significant underutilization problem, with an original compliance rate of only 27.35%. The optimization approach was 100% effective in addressing non-compliance, resulting in a net increase of nearly 7.5 million personnel hours across the system. This reveals an important insight: the "cost of compliance" to correct dangerous underutilization greatly surpassed any possible savings from eliminating overutilization, suggesting that the model is more than just a cost-saving tool; it is also an effective diagnostic tool. Finally, this project provides a robust, data-driven decision-support tool for healthcare administrators that replaces inefficient, manual scheduling with a mathematical approach to balancing operational efficiency and regulatory requirements, with future enhancements such as cost optimization or model deployment in a user-friendly dashboard.

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